

4.HISTOGRAM EQUALIZATION

AIM:

To enhance the contrast of a grayscale image using histogram equalization.

HISTOGRAM EQUALIZATION:

Histogram Equalization is a fundamental technique in digital image processing used to enhance the contrast of an image. It redistributes the intensity values of pixels so that they span the entire range of possible values more uniformly. This is particularly useful for images that are either too dark, too bright, or have low contrast.

1. Histogram Basics

A **histogram** of a grayscale image represents the frequency of occurrence of each intensity level within the image. For an image with L possible intensity levels $[0, L-1]$, the histogram is defined as:

$$h(r_k) = n_k \quad h(r_k) = n_k$$

where:

- r_k is the k th intensity level,
- n_k is the number of pixels in the image with intensity r_k .

The **normalized histogram** is given by:

$$p(r_k) = \frac{n_k}{MN} \quad p(r_k) = \frac{n_k}{MN}$$

where $M \times N$ is the total number of pixels in the image, and $p(r_k)$ represents the probability of occurrence of intensity level r_k .

```
Code:
a=imread("pout.tif");
nk=zeros(1,256);
[r,c]=size(a);
n=r*c;
for i=1:r
    for j=1:c
        intensity=a(i,j);
        nk(intensity+1)=nk(intensity+1)+1;
    end
end
figure;plot(nk);
%pdf
pk=nk/n;
%cdf
cdf(1)=pk(1);
for k=2:256
    cdf(k)=pk(k)+cdf(k-1);
end
sk=round(cdf*255);
rk=0:255;
figure;plot(rk,sk);
```

```

%applying sk
for i=1:r
    for j=1:c
        intensity=a(i,j);
        b(i,j)=sk(intensity+1);
    end
end
subplot(1,2,1),imshow(a);
subplot(1,2,2),imshow(uint8(b));
map=gray(256);
imwrite(uint8(b), map, "hist1.png");

```

Input image:



Output image:



Making Histogram Equalization Code as Function:

Function Code:

```

function output=histogram(a)
nk=zeros(1,256);
[r,c]=size(a);
n=r*c;
for i=1:r
    for j=1:c
        intensity=a(i,j);
        nk(intensity+1)=nk(intensity+1)+1;
    end
end
%pdf
pk=nk/n;
%cdf
cdf(1)=pk(1);
for k=2:256
    cdf(k)=pk(k)+cdf(k-1);
end
sk=round(cdf*255);
rk=0:255;

```

```
%applying sk
for i=1:r
    for j=1:c
        intensity=a(i,j);
        b(i,j)=sk(intensity+1);
    end
end
output=uint8(b)
```

Function Call:

```
f=imread("trees.tif");
h=histogram(f);
subplot(1,2,1),imshow(f);
subplot(1,2,2),imshow(h);
```

Input image:



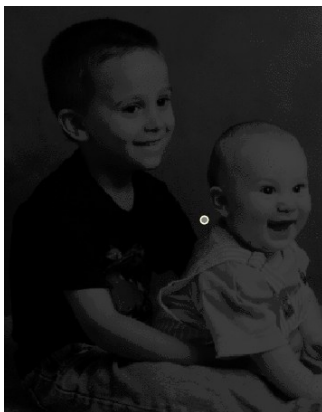
output image:



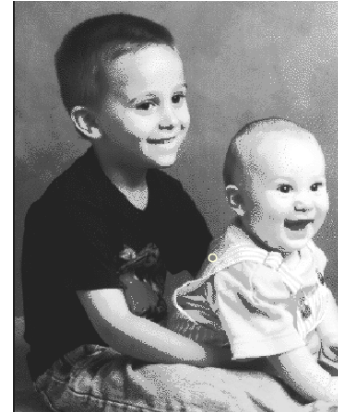
Histogram Equalization Using Inbuild Function:

```
a=imread("kids.tif");
b=histeq(a);
subplot(1,2,1),imshow(a);
subplot(1,2,2),imshow(uint8(b));
```

Input image:



output image:

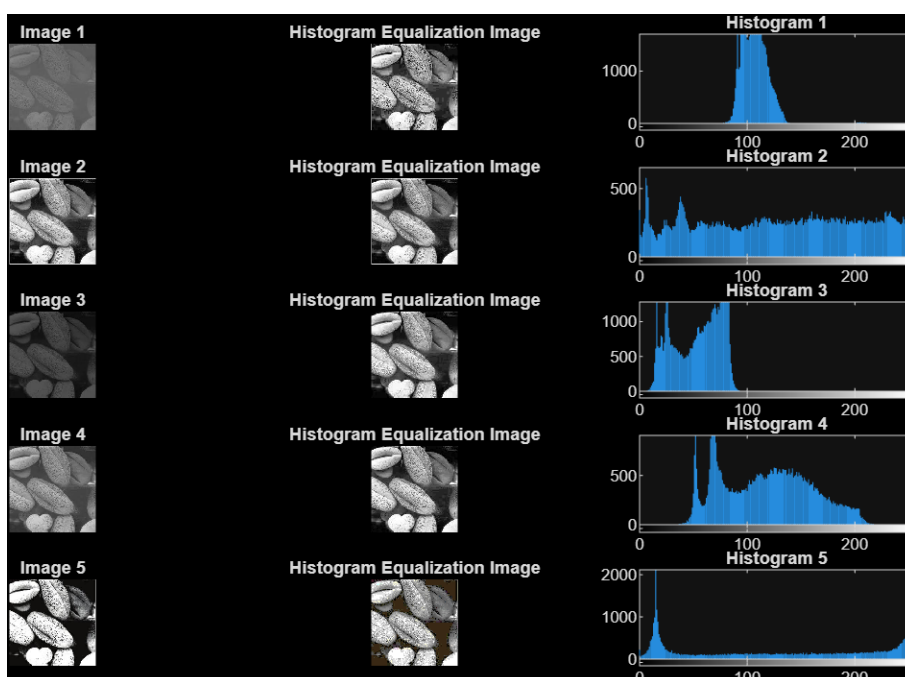


Code for rhist example:

Code:

```
a = imread("C:\Users\OneDrive\Pictures\rhist\cof1.jpg");
c = imread("C:\Users\OneDrive\Pictures\rhist\cof2.jpg");
e = imread("C:\Users\OneDrive\Pictures\rhist\cof3.jpg");
g = imread("C:\Users\OneDrive\Pictures\rhist\cof4.jpg");
j = imread("C:\Users\OneDrive\Pictures\rhist\cof5.jpg");
subplot(5,3,1), imshow(a), title('Image 1');
subplot(5,3,2), imshow(histeq(a)), title('Histogram Equalization Image');
subplot(5,3,3), imhist(rgb2gray(a)), title('Histogram 1');
subplot(5,3,4), imshow(c), title('Image 2');
subplot(5,3,5), imshow(histeq(c)), title('Histogram Equalization Image');
subplot(5,3,6), imhist(rgb2gray(c)), title('Histogram 2');
subplot(5,3,7), imshow(e), title('Image 3');
subplot(5,3,8), imshow(histeq(e)), title('Histogram Equalization Image');
subplot(5,3,9), imhist(rgb2gray(e)), title('Histogram 3');
subplot(5,3,10), imshow(g), title('Image 4');
subplot(5,3,11), imshow(histeq(g)), title('Histogram Equalization Image');
subplot(5,3,12), imhist(rgb2gray(g)), title('Histogram 4');
subplot(5,3,13), imshow(j), title('Image 5');
subplot(5,3,14), imshow(histeq(j)), title('Histogram Equalization Image');
subplot(5,3,15), imhist(rgb2gray(j)), title('Histogram 5');
```

Output image:



CONCLUSION:

Histogram equalization enhances image contrast by spreading out intensity values. It improves visibility in low-contrast images. Both manual and built-in methods were applied, showing clearer and more balanced output images. This technique is simple and effective for image enhancement.